1st: Aspect 9 (Understand the relationship between context and representation...)

- **Analysis:** This is the core competency required. The student must understand the logical mathematical relationship between the "real-world garage (Context)" and the "2D perspective drawings (Representation)." To solve the problem, one must use spatial reasoning to determine which mathematical representation (Options A–D) is "feasible" given the specific situational change (viewing from the back). It involves mapping an abstract geometric model to specific spatial orientations. (p. 1)

2nd: Aspect 5 (Interpret mathematical outcomes in a variety of formats... compare or evaluate...)

- **Analysis:** The problem provides "outcomes" in multiple formats (the four different back-view illustrations: A, B, C, and D). The student must compare and evaluate these different representations against the initial setup of George's garage to judge which one is correct. This requires a mental transformation and a comparative analysis of visual outputs. (p. 1)

3rd: Aspect 2 (Identify whether a... conclusion does, or does not, make sense given the context)

- **Analysis:** During the filtering process, the student performs logical checks. For example: "If the window is on the right-hand wall from the front, where should it appear when looking from the back?" The student must identify which conclusions (views) are "senseless" in terms of spatial logic (e.g., the window moving to the back wall or appearing on the wrong side) to eliminate incorrect options. (p. 1)

4th: Aspect 7 (Construct and communicate explanations and arguments...)

- **Analysis:** Although this is a multiple-choice question, the internal process involves constructing a rigorous "spatial argument." For instance: "The door is at the front, and the window is on the right; as I move to the back, the original right wall becomes the observer's left-hand side..." This internal logical derivation is the essence of constructing a mathematical explanation/argument. (p. 1)

第一位：方面 9 (Understand the relationship between context and representation...)

- **分析：** 这是本题最核心的考点。学生必须理解“现实情境中的车库（Context）”与“多个 2D 透视图（Representation）”之间的数学逻辑关系。学生需要利用对三维空间的理解，来解释在特定情境（背面视角）下，哪种数学表达（选项 A-D）是可行的。这涉及到将抽象的几何模型与具体的空间方位相匹配。

第二位：方面 5 (Interpret mathematical outcomes in a variety of formats... compare or evaluate...)

- **分析：** 题目提供了多种“表达格式”（A、B、C、D 四种不同的背面透视图）。学生必须通过评估和比较这些不同的表示形式，来判断哪一个符合题干中的初始设定。这要求学生在脑海中进行“视觉输出”的转换与对比。

第三位：方面 2 (Identify whether a... conclusion does, or does not, make sense given the context)

- **分析：** 在筛选选项时，学生需要进行逻辑判断。例如：“如果窗户在正视图的右侧墙壁上，那么从背面看时，如果能看到右侧墙壁，窗户应该出现在什么位置？”学生需要识别哪些结论（视图）在空间逻辑上是不合理的（如窗户位置移动到了后墙，或出现在了错误的侧面），从而排除错误选项。

第四位：方面 7 (Construct and communicate explanations and arguments...)

- **分析：** 虽然这是一个选择题，但解题过程需要学生在内心构建一套严密的“空间论证”。例如：“车库门在正面，窗户在右侧；当我转到背面时，原本的右侧墙变成了观察者的左侧方向.....”这种内部的逻辑推导正是构建数学解释和论据的过程。